Magnetic Induction Stirrer

Strong, motor-less and maintenance free

5 Year Warranty

Superior Functional Design

- Advanced magnetic coil technology drives the spinning of the stir bar maintaining continuous, constant speed for reproducible results
- Magnetic coupling is optimized by using alternating current (AC) as its driving force strong enough to stir viscous solutions
- Motor-less operation generates minimal heat ideal for stirring temperature sensitive samples and environments for long periods
- With no moving parts, produces less vibration and quieter operation over traditional magnetic stirrers

Intuitive Operation

- Multiple stirring modes are easy to program
- Bright digital display precisely controls time and speed
- Quickly meets stirring speeds with rapid acceleration and braking (less than 10 seconds)



Durable ABS housing is corrosion resistant and easy to clean





Item No.	Color	L x W x H *in*	L x W x H *cm*
120584	Grey/Purple	9.7 x 7.7 x 3	24.5 x 19.5 x 7.5
120585	Grey/Blue		

Replacement parts	
120738	MI Stirrer Silicone Mat
120739	MI Stirrer Universal Power Adapter

Specifications	
Speed Range	50–2,000 rpm variable, 1 rpm increments
Run Time Range	30 seconds to 60 minutes, variable, 1 second increments or continuous
Stirrer Modes	Single clockwise
	Single counter clockwise
	Auto reverse rotation 1 to 99 cycles 50% duty
Capacity	3 L plus
Mixing Surface	165 mm diameter, silicone mat 160 mm diameter

Package Includes
Rare Earth, NdFeB Magnetic Induction Stirrer
2 Stir Bars 35 mm x 12 mm Cylindrical Shape
1 Silicone Anti-Skid Plate Cover
1 Low Voltage, Double Insulated Power Adapter (100-240 VAC, 50/60 Hz) with 4 Interchangeable Plugs

Mini Magnetic Stirrer

Affordable solution, ultimate function

The Mini Magnetic Stirrer is an affordable, dependable solution ideal for general research, academic, life sciences, industrial and clinical lab settings

Durable Design

- Constructed of chemical resistant ABS and PC blend
- Compact footprint saves valuable space on benchtop
- Lightweight for easy transport

Ultimate Function

- Powerful magnetic coupling ensures accurate alignment, reducing spin-outs during acceleration or deceleration
- Performs routine stirring applications with low vibration and quiet operation



5 Year Warranty

U.S.and E.U. Utility Patent Pending
U.S. Design Patent No. D814,045

Item No.	Color	L x W x H *in*	L x W x H *cm*
120155	Grey/Purple	7.2 x 5.2 x 1.85	18.3 x 13.2 x 4.7
120595	Grey/Blue		

Replacement Part	
120815	Mini Stirrer Universal Power Adapter

Specifications	
Speed Range	200–2,200 rpm (Infinitely adjustable)
Capacity	Up to 3 L

Package Includes
Mini Magnetic Stirrer
2 Stir Bars 35 mm x 9 mm, Octagonal Shape
1 Low Voltage, Double Insulated Power Adapter (100-240 VAC, 50/60 Hz) with 4 Interchangeable Plugs



Stir Bars & Stir Bar Retriever

- Magnetic bar is made from ALNICO, one of the strongest permanent magnets available
- The sintered manufacturing process reduces the chance of decoupling (spin outs), encapsulation in PTFE protects both magnet and what it is stirring
- Chemically inert and easy to clean
- Can withstand temperatures up to 250°C, making them suitable for use with both unheated and heated stirrers
- Available in two surfaces:
  - Low friction octagon shape with a center pivot ring helps create a strong mixing vortex, even with more viscous solutions
  - Smooth shape for better coupling when using concave bottom glassware
- The retrieval bar can recover stir bars even in hot solutions when stirring and mixing is no longer required

Stir Bars

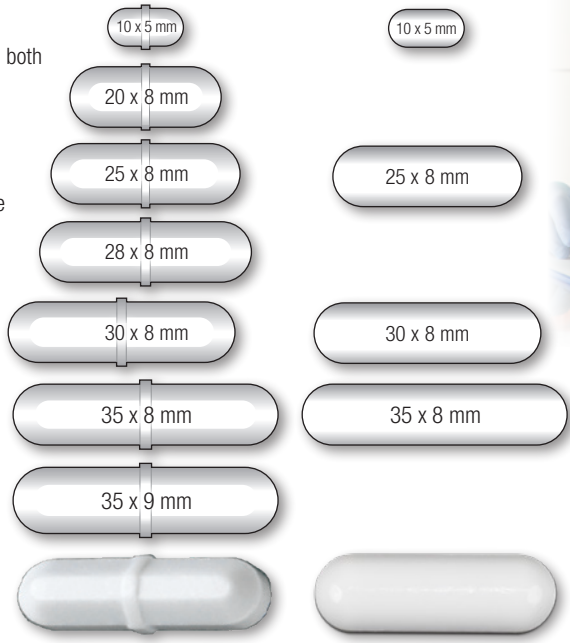
Pack of 10

Item No.	Shape	L x W *(in)*	L x W *(mm)*
120547	Octagonal with pivot ring	0.4 x 0.2	10 x 5
120548		0.8 x 0.3	20 x 8
120549		1 x 0.3	25 x 8
120550		1.1 x 0.3	28 x 8
120551		1.2 x 0.3	30 x 8
120552		1.4 x 0.4	35 x 9
120894	Smooth	0.4 x 0.2	10 x 5
120895		1 x 0.3	25 x 8
120896		1.2 x 0.3	30 x 8
120897		1.4 x 0.3	35 x 8

Stir Bar Retriever

Pack of 5

Item No.	L *(in)*	L *(cm)*
120553	11.8	30



Octagonal Shape

Smooth Shape



Stir Bar Retriever

NEW

5 Year Warranty

Digital Vortexer

Controlled, high-speed touch mixing

The Digital Vortexer combines digital precision, ergonomic comfort, and durable construction, featuring variable speed control, low activation force, high-speed mixing, stability, chemical resistance, compliance certifications and a 5-year warranty



Digital Control: produces more accurate results

- Digital controls enhance user control for precise mixing
- Illuminated digital display ensures accuracy, reproducibility, and consistent results
- Variable speed control offers flexibility and precise control of mixing intensity
- Ergonomic: prevents user fatigue
- Minimal activation force, supporting both touch start and continuous operation for convenience
- High-speed vortex capability – up to 3,600 RPM
- Multiple-position head allows vortexing at angles beyond 90 degrees for enhanced versatility
- Durable Design: guarantees quality that saves money
- Constructed from ABS and silicone, unit resists common chemicals and facilitates easy cleaning
- Stable platform and elastomer feet ensure unit stays stable during operation
- Meets Safety and Compliance Certifications while offering lower energy consumption
- 5-Year warranty



Item No.	Color	L x W x H *in*	L x W x H *cm*
120805	Grey/Purple	7.9 x 6.1 x 5.6	20.1 x 15.5 x 14.2
120806	Grey/Blue		

Digital Vortexer Accessories	
120922	Digital Vortexer Universal Power Adapter
120924	Digital Vortexer Replacement Silicone Cap

Specifications	
Operating Modes	Intermittent (Touch) and Continuous
Intermittent Speed Range	600 to 3,600 RPM
Continuous Speed Range	600 to 2,600 RPM
Run Time Range	0 to 50 hrs
Timer Range:	Down-counting, in minutes and seconds, (Continuous Mode run time is up to 50 hours, Intermittent Mode run time is up to 1 minute), Upward-counting run time is up to 50 hrs
Weight	5.4 lbs (2.4 kg)
Electrical Input	100-240 VAC, 2.5A, 50/60 Hz I Output: 12 VDC, 3.0A, 36.0W
Maximum Load (Continuous)	250 grams (0.55 lbs)



Models 120209, 120212, 120214

Vortexer

Multi-purpose, maximum convenience

Innovative Design

- One mixing head does it all with this sleek, modern and compact designed vortex mixer
- Hold slot configurations for multiple sizes of tubes, vessels and assay plates
- Center pad for quick hand-held mixing

Easy to Use

- Control switch with three operation selections (intermittent, off, continuous)
- Variable Speed Rotary Dial provides pre-set speed control selection during shaking mode

Durable Construction

- ABS material resist most commonly used chemicals and is easy to clean
- Stable elastomer feet dampen vibration and prevent movement during operation


Center pad area for hand held mixing


Assay plate x 1 (to be used with retention cords)


25 mL, 50 mL conical tubes x 2 (horizontally mounted)


15 mL conical tubes x 2 (horizontally mounted)


5 mL tubes x 2 (horizontally mounted in the same retention space as 15 mL)


0.5 mL microfuge tubes x 6 (vertically mounted)


1.5/2.0 mL microfuge tubes x 4 (vertically mounted)


0.2 mL microfuge tubes (strip tubes) x 24 (vertically mounted)

US Design Patent No. D787,085
US Utility Patent No. 9895670

Item No.	Speed Control	Power Input	L x W x H *in*	L x W x H *cm*
120209	Infinity Dial	110-120 VAC, 50/60 Hz, 55 W, US/NA	6.7 x 7.5 x 7.8 (with mixing head in place)	17.1 x 19 x 19.7 (with mixing head in place)
120212	Infinity Dial	220-240 VAC, 50 Hz, 44 W, CE		
120214	Infinity Dial	220-240 VAC, 50 Hz, 44 W, UK		
120210	Preset Dial	220-240 VAC, 50 Hz, 44 W, AUS		
120318	Preset Dial	100-110 VAC, 50/60 Hz, 66 W(PSE), Japan		

Replacement parts	
120327	Vortexer Multi-Head
120328	Vortexer Central Pad, *3 pk*
120422	Vortexer Retention Cords, *3 pk*
120329	100-120 VAC US Cord

Replacement parts	
120333	220-240 VAC AUS Cord
120332	220-240 VAC CE Cord
120331	220-240 VAC UK Cord
120330	100-110 VAC Japan Cord

Specifications:	
Speed Range	Up to 3,000 rpm
Orbit	3.7 mm
Control	3 Position Control — (top) intermittent, (middle) off, (bottom) continuous



Mini Vortexer

Small footprint with maximum power

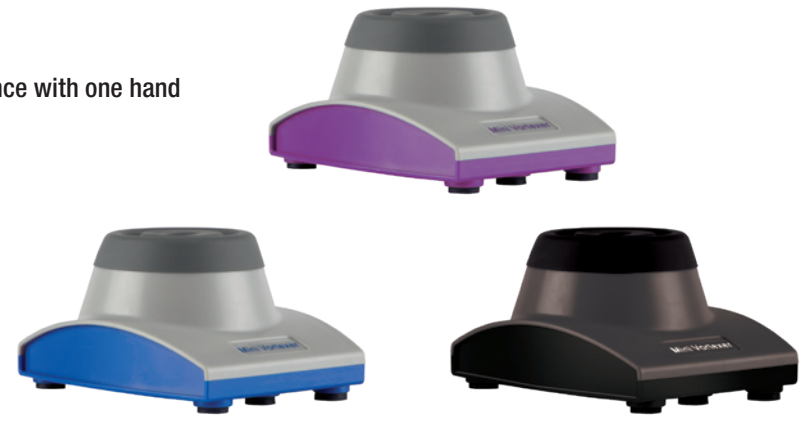
Simple, touch start operation offers exceptional mixing performance with one hand

Robust Design

- Low profile design and small footprint saves valuable space
- Made of ABS and silicone materials that resist most commonly used chemicals
- Stable platform and elastomer feet ensure unit stays stable during operation

Powerful Function

- Rapid vortex action with low activation force
- Quiet operation and low vibration



U.S. and E.U. Utility Patent Pending
U.S. Design Patent No. D814,045

Item No.	Color	L x W x H *in*	L x W x H *cm*
120567	Grey/Purple	5.1 x 4.3 x 3	13 x 11 x 7.5
120598	Grey/Blue		
120827	Black/Black		

Replacement parts	
120617	Mini Vortexer Silicone Cover, set of 2
120814	Mini Vortexer Universal Power Adapter

Specifications	
Speed	4,500 rpm
Orbit	6 mm
Capacity	Up to 50 mL tube

Package Includes
Mini Vortexer
2 Silicone Caps
1 Low Voltage, Double Insulated Power Adapter (100-240 VAC, 50/60 Hz) with 4 Interchangeable Plugs



Digital Orbital Shaker

Shaker for a vast range of protocols

Versatile Design

- Create your own matrix for vessels of all shapes and sizes with Hi-Lo Platform and Bungee Cords
- Instantly adjusts to weight load changes and unbalanced loads with no-stop electronic motor feedback allowing for truly reproducible results
- Durable ABS construction suitable for use in incubators (excluding C02) and cold rooms within the stated temperature range

High-Featured Operation

- Suitable for wide range of protocols including very low start speed for sensitive sample agitation to high speeds for vigorous applications
- Soft start and shut down feature reduces spilling from full vessels
- Designed to handle heavy loads up to 10 lbs (4.5 kg)
- Overload protection shut-off prevents overloading



Item No.	L x W x H *in*	L x W x H *cm*	Platform Interior W x D *in*	*cm*
120460	12.4 x 7.8 x 12.2	31.4 x 19.7 x 31	11 x 11	27.9 x 27.9

Replacement parts	
120521	Digital Orbital Shaker Silicone Mat & Bumpers
120524	Digital Orbital Shaker Bungee Cords, *Pack of 8*
120491	Digital Orbital Shaker Universal Power Adapter

Specifications	
Speed Range	20–300 rpm, 10 rpm increments
Orbit Diameter	19 mm (0.8")
Run Time Range	0 to 74 hours and 59 minutes; 1 minute increments or continuous
Maximum Load	4.5 kg (10 lbs)

Package Includes
Digital Orbital Shaker
Skid-Free Rubber Platform Mat
8 Bungee Cord Set
Universal Hi-Lo Platform with Soft Rubber Edge Bumpers
1 Low Voltage, Double Insulated Power Adapter (100-240 VAC, 50/60 Hz) with 4 Interchangeable Plugs

Sampling & Testing Instruments

Reliable and accurate instrument range for diverse applications



Thermometers

Refractometers

Hydrometers & Jars

Pipette Controllers

Colony Counters & Accessories

THERMOMETERS

We have a wide range of thermometers to measure and track temperatures for a wide range of applications.
Our portfolio includes Dial (Bimetal), Digital and Glass Thermometer options to get the job done.

Dial Thermometers

Dial thermometers don't require batteries, are enduring, and readings can be adjusted with the calibration screw on the back

Indicates temperature through a metal stem with attached circular scale.
Application use varies with dial sizes, stem lengths and temperature ranges.



Dial Thermometers - 1" Diameter

Ideal for many applications including food and beverage quality control, general lab use, and asphalt/concrete testing

- The small stem and face fits nicely into a shirt pocket or holder
- Includes a plastic sheath with pocket clip for storage

Dial Thermometers - 1" Diameter

Item No.	Temperature Range	Subdivisions	Accuracy	Face Diameter *(in/mm)*	Stem Length *(in/mm)*	Stem Diameter *(in/mm)*
81125	25 to 125°F	1°F	±1°F	1 / 25	5 / 127	0.1 / 3.5
81070	-40 to 70°C	1°C	±1°C			
81110	-10 to 110°C	1°C	±1°C			
81550	50 to 550°F	5°F	±5°F			



Dial Thermometers - 1.75" / 2" Diameter

Ideal for many applications including food and beverage quality control, general lab use, and asphalt/concrete testing

- Larger dial size than common pocket thermometers for easier reading
- Features an adjustment screw for field calibration
- Includes a clip for container mounting

Dial Thermometers - 1.75" / 2" Diameter

Item No.	Temperature Range	Subdivisions	Accuracy	Face Diameter *(in/mm)*	Stem Length *(in/mm)*	Stem Diameter *(in/mm)*
82125	25 to 125°F	1°F	±1°F	1.8 / 44.5	8 / 203	0.1 / 3.5
82110	-10 to 110°C	1°C	±1°C	1.8 / 44.5	8 / 203	0.1 / 3.5
82220	0 to 220°F	2°F	±2°F	1.8 / 44.5	8 / 203	0.1 / 3.5
82550	50 to 550°F	5°F	±5°F	1.8 / 44.5	8 / 203	0.1 / 3.5
82550DG	50 to 550°F / 10 to 290°C	2.5°F	±2.5°F	2 / 51	8 / 203	0.1 / 3.5
		5°C	±5°C			

Dial Thermometers - 3" Dial

Ideal for many applications including soil testing, food and beverage quality control, and asphalt/concrete testing

- Constructed of stainless-steel case, bezel, and stem with a glass window
- Case is hermetically sealed to prevent lens fogging
- Includes a 1/2"NPT male thread on the back for threading into a pipeline
- Also features an adjustment screw for field calibration



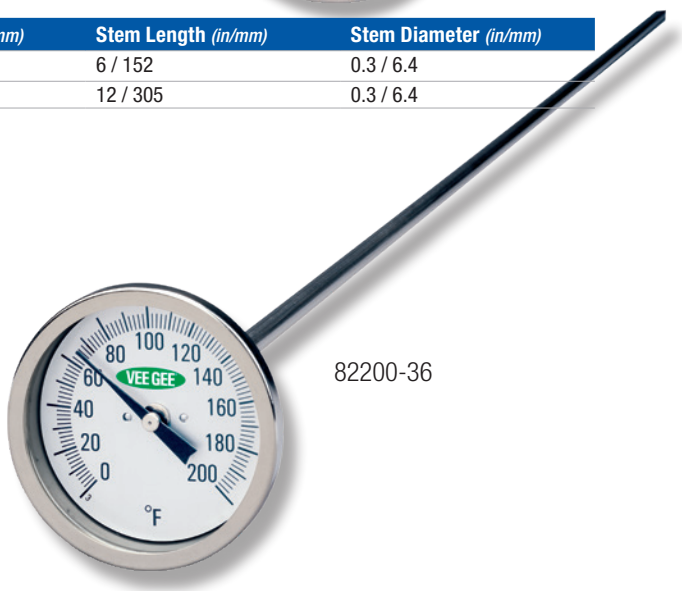
Dial Thermometers - 3" Diameter

Item No.	Temperature Range	Subdivisions	Accuracy	Face Diameter *(in/mm)*	Stem Length *(in/mm)*	Stem Diameter *(in/mm)*
82160-6	-40 to 160°F	2°F	±2°F	3 / 76	6 / 152	0.3 / 6.4
82160-12	-40 to 160°F	2°F	±2°F	3 / 76	12 / 305	0.3 / 6.4

Long-Stem Dial Thermometers

Ideal for many applications including compost testing, food and beverage quality control, and asphalt/concrete testing

- Constructed of stainless-steel case, bezel, and stem with a glass window
- Case is hermetically sealed to prevent lens fogging
- Includes a 1/2"NPT male thread on the back for threading into a pipeline
- Also features an adjustment screw for field calibration



Long-Stem Dial Thermometers

Item No.	Temperature Range	Subdivisions	Accuracy	Face Diameter *(in/mm)*	Stem Length *(in/mm)*	Stem Diameter *(in/mm)*
82200-36	0 to 200°F	2°F	±2°F	3 / 76	36 / 914	0.3 / 7.9
82100-36	0 to 100°C	1°C	±1°C	3 / 76	36 / 914	0.3 / 7.9
82200-48	0 to 200°F	2°F	±2°F	3 / 76	48 / 1220	0.3 / 7.9

Swivel-Head Digital Thermometers

Digital stem thermometers are ideal for temperature monitoring in essentially any setting and application

- Laboratories - easily monitoring reagents and liquids in test tubes, beakers, and flasks
- Food Processing
- Refrigeration
- Horticulture

Both models feature a dual scale (°F / °C), fast 1-second update time, and a swivel head that can be moved a full 180° for easy reading in any position. Both the ON/OFF and scale selection buttons are located on the face for easy access.

The thermistor sensor stem is stainless steel, 0.138" (3.5 mm) diameter, and includes a protective case featuring a pocket clip. Includes a 1.5V alkaline battery (LR44) and features automatic power-off after 10 minutes of non-use.



Swivel-Head Digital Thermometers

Item No.	Temperature Range	Resolution	Accuracy	Stem Length *(in/mm)*	Stem Diameter *(in/mm)*
83210-12	-58 to 302°F / -50 to 150°C	0.1°	±2.0°F / 1.0°C	12 / 305	0.1 / 3.5
83110	14 to 212°F / -10 to 100°C	0.1°	±1.0°F / 0.5°C	5 / 127	0.1 / 3.5

Glass Thermometers

VEE GEE thermometers feature a SafetyBLUE fill. This is an organic performance liquid that is bio-degradable and exhibits all the qualities of mercury without the risk of contamination and hazards associated with mercury.

Serialized Glass Thermometers

Precision serialized thermometers are ideal for use in laboratories, universities, food/beverage, environmental, wastewater and many other general-purpose applications

- Feature SafetyBLUE fill against a white background for easy reading
- Accuracy is equal to one subdivision

Serialized Glass Thermometers

Item No.	Temperature Range	Subdivisions	Length (in/mm)	Immersion
80701	-30 to 120°F	2°F	12 / 300	Total
80702	-4 to 230°F	2°F	12 / 300	Total
80703	0 to 300°F	2°F	12 / 300	Total
80706	-35 to 50°C	1°C	12 / 300	Total
80707	-20 to 110°C	1°C	12 / 300	Total
80708	-20 to 150°C	1°C	12 / 300	Total
80901	-30 to 120°F	2°F	12 / 300	Partial
80902	-4 to 230°F	2°F	12 / 300	Partial
80903	0 to 300°F	2°F	12 / 300	Partial
80905	20 to 500°F	2°F	15.4 / 391	Partial
80906	-35 to 50°C	1°C	12 / 300	Partial
80907	-20 to 110°C	1°C	12 / 300	Partial
80908	-20 to 150°C	1°C	12 / 300	Partial
80910	-10 to 250°F	2°F	15.4 / 391	Partial



80901

Serialized Encapsulated Glass Thermometers

Encapsulation protects you from broken glass and liquid

- Feature FEP encapsulation; in the event of breakage the FEP coating contains the broken glass and liquid, preventing injury or contamination
- Feature SafetyBLUE fill against a white background for easy reading
- Accuracy is equal to one subdivision

Serialized Encapsulated Glass

Item No.	Temperature Range	Subdivisions	Length *(in/mm)*	Immersion
80701E	-30 to 120°F	2°F	12 / 300	Total
80702E	-4 to 230°F	2°F	12 / 300	Total
80703E	0 to 300°F	2°F	12 / 300	Total
80704E	30 to 220°F	2°F	15.4 / 391	Total
80706E	-35 to 50°C	1°C	12 / 300	Total
80707E	-20 to 110°C	1°C	12 / 300	Total
80708E	-20 to 150°C	1°C	12 / 300	Total
80901E	-30 to 120°F	2°F	12 / 300	Partial
80902E	-4 to 230°F	2°F	12 / 300	Partial
80903E	0 to 300°F	2°F	12 / 300	Partial
80904E	30 to 220°F	2°F	15.4 / 391	Partial
80906E	-35 to 50°C	1°C	12 / 300	Partial
80907E	-20 to 110°C	1°C	12 / 300	Partial
80908E	-20 to 150°C	1°C	12 / 300	Partial



80702E

Serialized Armored Glass Thermometers

Armor protects the thermometer from rough handling and shock during use

- Includes nickel-plated brass armor
- Feature SafetyBLUE fill against a white background for easy reading
- Accuracy is equal to one subdivision

Serialized Armored Glass Thermometers

Item No.	Temperature Range	Subdivisions	Length *(in/mm)*	Immersion
80701-A	-30 to 120°F	2°F	12 / 300	Total
80702-A	-4 to 230°F	2°F	12 / 300	Total
80703-A	0 to 300°F	2°F	12 / 300	Total
80706-A	-35 to 50°C	1°C	12 / 300	Total
80707-A	-20 to 110°C	1°C	12 / 300	Total
80708-A	-20 to 150°C	1°C	12 / 300	Total
80901-A	-30 to 120°F	2°F	12 / 300	Partial
80902-A	-4 to 230°F	2°F	12 / 300	Partial
80903-A	0 to 300°F	2°F	12 / 300	Partial
80905-A	20 to 500°F	2°F	15.4 / 391	Partial
80906-A	-35 to 50°C	1°C	12 / 300	Partial
80907-A	-20 to 110°C	1°C	12 / 300	Partial
80908-A	-20 to 150°C	1°C	12 / 300	Partial
80910-A	-10 to 250°F	2°F	15.4 / 391	Partial



80701-A

Armored Encapsulated Glass Thermometers

Armor protects the thermometer and encapsulation protects you

- Includes FEP encapsulation and nickel-plated brass armor
- In the event of breakage, the FEP coating contains the broken glass and liquid fill, preventing injury or contamination. The armor protects the thermometer from rough handling and shock during use
- Feature SafetyBLUE fill against a white background for easy reading
- Accuracy is equal to one subdivision

Armored Encapsulated Glass Thermometers

Item No.	Temperature Range	Subdivisions	Length *(in/mm)*	Immersion
80701E-A	-30 to 120°F	2°F	12 / 300	Total
80702E-A	-4 to 230°F	2°F	12 / 300	Total
80703E-A	0 to 300°F	2°F	12 / 300	Total
80704E-A	30 to 220°F	2°F	15.4 / 391	Total
80706E-A	-35 to 50°C	1°C	12 / 300	Total
80707E-A	-20 to 110°C	1°C	12 / 300	Total
80708E-A	-20 to 150°C	1°C	12 / 300	Total
80901E-A	-30 to 120°F	2°F	12 / 300	Partial
80902E-A	-4 to 230°F	2°F	12 / 300	Partial
80903E-A	0 to 300°F	2°F	12 / 300	Partial
80904E-A	30 to 220°F	2°F	15.4 / 391	Partial
80906E-A	-35 to 50°C	1°C	12 / 300	Partial
80907E-A	-20 to 110°C	1°C	12 / 300	Partial
80908E-A	-20 to 150°C	1°C	12 / 300	Partial

Armor Cases for Glass Thermometers

Nickel-plated brass armor cases protect liquid-in-glass thermometers from rough handling and shock during use

Armors feature a threaded screw cap with ring top for suspension where necessary

Item No.	Thermometer Length *(in/mm)*	Outside Diameter *(in/mm)*	Inside Diameter *(in/mm)*
83025A	12 / 305	0.4 / 9.5	0.3 / 7.5
83025N	15.5 / 393	0.4 / 9.5	0.3 / 7.5



83025A

Glass Thermometers: Be in the know....

Column Separation

The largest single cause for failure of liquid-in-glass thermometers in the lab is due to column separation. This can occur in transit or in the lab.

The simplest and safest method to reunite a separated column is to force the liquid down the capillary tube by using a centrifuge, if one is available with a cup deep enough to ensure that the centrifugal force is below the liquid column.

- Carefully insert the thermometer, bulb down, into the centrifuge. Place some cotton wadding at the bottom of the cup to prevent any damage to the bulb. Turn on the centrifuge - in just a few seconds all the liquid will be forced past the separation.
- If the cup is not deep enough and all the centrifugal force is not below the column, the column will split, forcing part of the liquid down. The remainder will be forced up, filling the expansion chamber.
- If a centrifuge is not available, the column can be reunited by holding the thermometer in an upright position and gingerly tapping the stem above the separation against the palm of your hand. As you gently tap the thermometer, observe the liquid above the separation until it breaks away from the wall of the capillary and runs down to join the main column.

Immersion Types

Thermometers are one of the most important instruments used in the laboratory and are a relatively inexpensive means of reliably measuring temperature, provided they are properly used and not abused.

It is essential that the instruments be used in the manner prescribed if true temperature are to be ascertained. The following are some terms that should be understood and adhered to when true temperatures are to be obtained:

Total immersion - such a thermometer is designed to indicate temperatures correctly when the bulb and the entire liquid column are exposed to the temperature being measured, except for a minimal length emergent to be visible. Therefore, the thermometer is immersed into the medium up to the point of reading.

Partial Immersion - such a thermometer has a line around it at the immersion distance from the bottom. The thermometer indicates correctly when the bulb and liquid column to that line are exposed to the temperature being measured and the emergent stem is at ambient temperature. Therefore, the thermometer is immersed only up to the immersion line.

REFRACTOMETERS

VEE GEE's benchtop and handheld refractometers provide users with highly accurate readings for a wide range of applications in the agricultural, food & beverage, chemical and manufacturing industries.

We offer a large breadth of scales for specific applications such as Brix, NaCl, Salinity, Specific Gravity of Seawater, Coolant, Battery Acid, and more.

Our portable line of optical and digital refractometers offers the convenience to perform sample analysis where and when you need. With added temperature control, our Abbe benchtop unit takes the accuracy of your Brix/Refractive Index analysis to the next level in the laboratory.

X-Series Handheld Refractometers

Compact & Portable

The X-Series handheld refractometers are exceptionally simple to operate and provide quick, accurate measurements for the concentration of nearly any aqueous solution in a wide range of applications.

Made durable with industrial-grade design features to provide years of reliable use even in the field. Rugged metal housing construction with slip-resistant rubber grip and a Non-Roll stand to protect from rolling off surfaces.

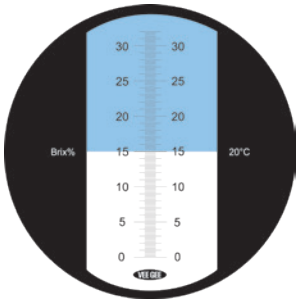
No need to recalibrate or apply correction factors in work environments with large temperature swings with the Automatic Temperature Compensation (ATC) feature included on Models BTX-1, BTX-20, STX-3, CTX-1 & CTX-2. The effective compensation range is 50-86˚ F / 10-30˚ C.

5 Year Warranty



Protective Grip
The rubber grip reduces hand slipping and guards against user's hand heat to maintain accuracy

Tool-Free Calibration
Tools not required with our patented "Zero-Ring" calibration (non ATC units only)



Accurate & Sharp Readings
Large scale is best in class delivering clear, sharp figures, screen shading and incredible contrast for overall accuracy

Quick Results
High quality optical glass prism and all-metal housing design allows the sample and prism to reach temperature equilibrium quickly

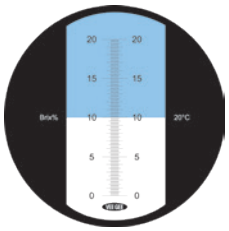
X-Series Specifications	
Sample Type	Transparent and Translucent
Sample Quantity	>0.1 mL
Prism	Optical Glass
Prism Housing	Copper Alloy w/Nickel-Chrome Coating
Included	Carrying Case, Plastic Transfer Pipet, Instruction Manual * ATC models include calibration screwdriver.
Warranty	5-Year

X-Series Handheld Refractometers

For Food, Beverage, and Agriculture Applications

Brix Handheld Refractometers

Item No.	Model No.	Scale	Range	Resolution	Accuracy	ATC	Approx Length *(in/mm)*
43001	BX-1	Brix	0 to 32%	0.2%	±0.2% Brix	No	6.5 / 165
43002	BTX-1	Brix	0 to 32%	0.2%	±0.2% Brix	Yes	6.5 / 165
43003	BX-2	Brix	28 to 62%	0.2%	±0.2% Brix	No	5.9 / 150
43004	SGX-1	Brix/Specific Gravity Beer Wort	0 to 32% 1.000 to 1.130	0.2% / 0.001	±0.2% / ±0.001	Yes	6.5 / 165
43005	BX-3	Brix	60 to 92%	0.2%	±0.2% Brix	No	5.5 / 140
43006	HMX-1	Brix, Honey Moisture	Brix, 58.0 to 92%	0.5%	±0.5% Brix	Yes	5.8 / 148
			Water, 13 to 27%	1.0%	±1.0% Water		
			Baume, 38 to 43˚	0.5˚	±0.5˚ Baume		
43007	BX-4	Brix	45 to 82%	0.2%	±0.2% Brix	No	5.5 / 140
43009	BX-10	Brix	0 to 10%	0.1%	±0.1% Brix	No	7.3 / 185
43011	BX-20	Brix	0 to 20%	0.1%	±0.1% Brix	No	7.3 / 185
43012	BTX-20	Brix	0 to 20%	0.1%	±0.1% Brix	Yes	7.3 / 185
43015	BX-50	Brix	0 to 50%	0.5%	±0.5% Brix	No	5.7 / 145



43012/Model BTX-20
For specific scale images, visit product on our website.



New! Brix Field Kits

Everything you need for in field testing at the Vineyard or Honey Bee Farm

Includes durable Carrying Case with Refractometer, Microfiber Cloth, Bottle for Distilled Water, Sample Cup, Sample Pipette, Calibration Screwdriver and Manual

Item No.	Model No.	Scale	Range	Resolution	Accuracy	ATC	Approx. Length *(in/mm)*	L x W x H of Kit *(in/cm)*
43002-KIT	BTX-1	Brix	0 to 32%	0.2%	±0.2% Brix	Yes	6.5 / 165	10.5 x 7 x 3 in. 26.7 x 17.8 x 7.6 cm
43006-KIT	HXM-1	Brix, Honey Moisture	Brix, 58.0 to 92%	0.5%	±0.5% Brix	Yes	5.8 / 148	
			Water, 13 to 27%	1.0%	±1.0% Water			
			Baume, 38 to 43˚	0.5˚	±0.5˚ Baume			

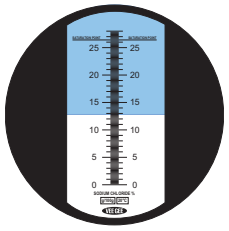


Water Testing Refractometers

For Water/Wastewater, Aquatic and Aquaculture Applications

Sodium Chloride And Salinity Handheld Refractometers

Item No.	Model No.	Scale	Range	Resolution	Accuracy	ATC	Approx.Length *(in/mm)*
43033	SX-1	NaCl	0 to 10%	0.1%	±0.1%	No	7.3 / 185
43035	SX-2	NaCl	0 to 28%	0.2%	±0.2%	No	6.5 / 165
43036	STX-3	Salinity	0 to 100 ppt	1.0 ppt	±1.0 ppt	Yes	7.3 / 185
		Specific Gravity of Seawater	1.000 to 1.070	0.001	±0.001		



43035/Model SX-2
For specific scale images, visit product on our website.



New! Water Testing Field Kit

Testing when and where you need to with this convenient kit

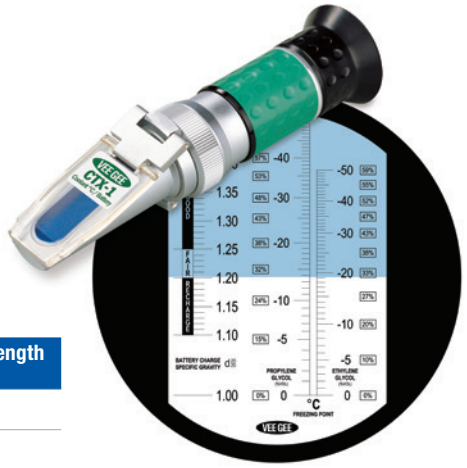
Includes durable Carrying Case with Refractometer, Microfiber Cloth, Bottle for Distilled Water, Sample Cup, Sample Pipette, Calibration Screwdriver and Manual

Item No.	Model No.	Scale	Range	Resolution	Accuracy	ATC	Approx. Length *(in/mm)*	L x W x H of Kit *(in/cm)*
43036-KIT	STX-3	Salinity	0 to 100 ppt	1.0 ppt	±1.0 ppt	Yes	7.3 / 185	10.5 x 7 x 3 in. 26.7 x 17.8 x 7.6 cm
		Specific Gravity of Seawater	1.000 to 1.070	0.001	±0.001			



X-Series Handheld Refractometers

For Light Industrial, Chemical, and Automotive Applications



43062/Model CTX-1
For specific scale images, visit product on our website.

General Purpose and Industrial Chemical Handheld Refractometers

Item No.	Model No.	Scale	Range	Resolution	Accuracy	ATC	Approx. Length *(in/mm)*
43055	NDX-2	Refractive Index	1.435 to 1.520	0.001	±0.001	No	5.5 / 140
43062	CTX-1	Propylene Glycol Freezing Point	-58 to 32°F / -50 to 0°C	33.80°F 1°C	±1°C	Yes	5.9 / 150
		Ethylene Glycol Freezing Point	-50 to 0°C	1°C	±1°C		
		Battery Acid Specific Gravity	1.10 to 1.40	0.01	±0.01		
43064	CTX-2	Propylene Glycol Freezing Point	-50 to 32°F / -45.56 to 0°C -60 to 32°F / -51.11 to 0°C	2°F -16.67°C	±2°F	Yes	5.9 / 150
		Ethylene Glycol Freezing Point	-60 to 32°F	2°F	±2°F		
		Battery Acid Specific Gravity	1.10 to 1.40	0.01	±0.01		
43071	URX-60	Urea	0 to 60%	0.5%	±0.5%	No	5.7 / 145

Accessories

Item No.	Description	Model
43000-DLXXX	Daylight Plate Kit	X-Series Handheld Refractometers Kit includes: Ring, Hinge, Plate, and Screw. Assembled
43RP-008	Soft Fabric Carrying Case	X-Series Handheld Refractometers
1111	Soft Vinyl Carrying Case	BX-1, BTX-1, BX-2, BX-3, BX-4, BX-50, SX-2, NDX-2, CTX-1, CTX-2, HMX-1, SGX-1, URX-60
1111L	Soft Vinyl Carrying Case	BX-10, BX-20, BTX-20, SX-1, STX-3



43000-DLXXX
Daylight Plate Kit

Transfer Pipettes

One-piece, unbreakable polyethylene plastic pipettes

Transfer and deliver small volumes of liquid quickly and safely

- Bulk packed versions come in handy dispenser box
- Sterile versions available single wrapped or 20 wrapped at a time
- Lot-to-lot traceability on sterile versions

STERILE R



Transfer Pipettes

	Item No.	Vol. (mL)	Working Vol. (mL)	Approx. Length *(in/mm)*	Graduations *(mL)*	Packaging 500 per box	Sterile
	HS206371A	5	1	5.9 / 150	0.25	Bulk	
a	**HS206371B**	5	1	5.9 / 150	0.25	20/bag	✓
	HS206371C	5	1	5.9 / 150	0.25	1/single wrap	✓
	HS206372A	5	2	6.1 / 155	0.5	Bulk	
b	**HS206372B**	5	2	6.1 / 155	0.5	20/bag	✓
	HS206372C	5	2	6.1 / 155	0.5	1/single wrap	✓
	HS206373A	7	3	6.1 / 155	0.5	Bulk	
c	**HS206373B**	7	3	6.1 / 155	0.5	20/bag	✓
	HS206373C	7	3	6.1 / 155	0.5	1/single wrap	✓

MDX-Series
Digital Refractometers

Compact, Durable and Accurate

The MDX-Series Digital Refractometers feature IP65/67 ratings, making them ideal for use in the field and in a multitude of environments.

Delivering quick digital accuracy with high-end performance features like:

- Automatic Temperature Compensation (ATC)
- Continuous Loop Average Readings



Intuitive to Use
Easy operation with large readable digital LCD display that shows all data needed on screen



Easy to Clean
Prism cover and rubber grip add extra protection and are removable for easy cleaning

Quick Results
Simply place 4-5 drops of your sample into the well, close the prism cover, and press the "READ" button



1 Year Warranty
Parts and Labor

MDX-Series Specifications	
ATC Range	32 to 104°F / 0 to 40°C
IP Rating	IP67: detecting unit IP65: body
Prism	Optical Glass
Wavelength	589.3 nm LED
Sample Cell	Stainless Steel
Calibration Solution	Distilled Water (not included)
Min. Sample Volume	0.3 mL
Battery	1 X AAA, 1.5V
Dimensions (L x W x H in/mm)	4.8 x 2.3 x 1.3 / 121.92 x 58.42 x 33.02
Weight (lb/kg)	0.198 lb / 0.089 kg
Protective Case Dimensions (L x W x H in/mm)	6.75 x 4.25 x 2 / 171.45 x 107.95 x 50.8
Included	Hard shell case with protective lining, disposable pipette, (1) AAA Battery, and user manual
Warranty	1-Year Parts and Labor

MDX-Series Digital Refractometers

For Food, Beverage, and Agriculture Applications

Digital Brix/Sugar Refractometer

Item No.	Model No.	Scale	Range	Resolution	Accuracy
48101	MDX-101	Brix	0.0 to 50.0% Brix	0.1% Brix	±0.2% Brix
		Refractive Index	1.3330 to 1.4200 nD	0.0001 nD	±0.0003 nD
48102	MDX-102	Brix	0.0 to 90.0% Brix	0.1% Brix	±0.2% Brix
		Refractive Index	1.3330 to 1.5300 nD	0.0001 nD	±0.0003 nD
48103	MDX-103	Dextran	0.0 to 10.5%	0.1%	±0.2%
		Fructose	0.0 to 68.0%	0.1%	±0.2%
		Glucose	0.0 to 59.0%	0.1%	±0.2%
		Lactose	0.0 to 16.5%	0.1%	±0.2%
		Maltose	0.0 to 15.5%	0.1%	±0.2%
		Sucrose (Brix)	0.0 to 90.0%	0.1%	±0.2%

Digital Honey Refractometer

Item No.	Model No.	Scale	Range	Resolution	Accuracy
48301	MDX-301	Brix	0.0 to 90.0% Brix	0.1% Brix	±0.2% Brix
		Moisture	5.0 to 38.0%	0.1%	±0.2%
		Baume	33.0 to 48.0°	0.1°	±0.2°
		Refractive Index	1.3330 to 1.5300 nD	0.0001 nD	±0.0003 nD

Digital Wine Refractometer

Item No.	Model No.	Scale	Range	Resolution	Accuracy
48401	MDX-401	Brix	0.0 to 50.0% Brix	0.1% Brix	±0.2% Brix
		PA (% v/v)	0.0 to 22.0%	0.1%	±0.2%
		Oechsle (Oe)	0 to 150°	1°	±2°
		KMW (Babo)	0.0 to 25.0°	0.1°	±0.2°

For Water/Wastewater, Aquatic and Aquaculture Applications

Digital Sodium Chloride Refractometer

Item No.	Model No.	Scale	Range	Resolution	Accuracy
48201	MDX-201	NaCl (%)	0.0 to 28.0%	0.1%	±0.2%
		NaCl (ppt)	0 to 280 ppt	1 ppt	±2 ppt
		Specific Gravity	1.000 to 1.217	0.001	±0.002
		Refractive Index	1.3330 to 1.4200 nD	0.0001 nD	±0.0003 nD

Digital Seawater Refractometer

Item No.	Model No.	Scale	Range	Resolution	Accuracy
48202	MDX-202	Salinity	0 to 100 ppt	1 ppt	±2 ppt
		Chlorinity	0 to 57 ppt	1 ppt	±2 ppt
		Specific Gravity	1.000 to 1.075	0.001	±0.002
		Refractive Index	1.3330 to 1.4200 nD	0.0001 nD	±0.0003 nD

For Light Industrial, Chemical, and Automotive Applications

Digital DEF (UREA)/Battery Acid Refractometer

Item No.	Model No.	Scale	Range	Resolution	Accuracy
48602	MDX-602	DEF (Urea)	0.0 to 51.0%	0.1%	±0.2%
		Refractive Index	1.3330 to 1.4200 nD	0.0001 nD	±0.0003 nD
		Battery Acid Specific Gravity	1.000 to 1.500	0.001	±0.005

Digital Coolant Refractometer

Item No.	Model No.	Scale	Range	Resolution	Accuracy
48603	MDX-603	Ethylene Glycol (% v/v)	0.0 to 65.0%	0.1%	±0.5%
		Ethylene Glycol Freezing Point	-80.0 to 32.0°F / -50.0 to 0.0°C	0.1°F / 0.1°C	±1.0°F / ±0.5°C
		Propylene Glycol (% v/v)	0.0 to 73.0%	0.1%	±0.5%
		Propylene Glycol Freezing Point	-100.0 to 32.0°F / -50.0 to 0.0°C	0.1°F / 0.1°C	±1.0°F / ±0.5°C

C10
Abbe Refractometer

High Precision Benchtop Unit

The C10 Abbe Refractometer is a high precision benchtop instrument measuring the Refractive Index (nD) of liquids, glass, plastic and film.

With an oversized, easy to read scale and Prism Temperature Control feature, the C10 delivers Brix and Refractive Index Scales and dispersion measurements with high precision.

Made of easy to clean, durable metal housing, the C10 is a long-lasting tool perfect for Education, Food, Beverage and Industrial Laboratories.

C10 Abbe Refractometer Specifications	
Dispersion (nF-nC)	Dial Readings w/Conversion Charts
Ambient Temperature	41 to 95°F / 5 to 35°C 32 to 158°F / 0 to 70°C
Measuring Temperature	0 to 70°C
Optical Wavelength	589 nm
Prism	Optical Glass
Prism Housing	SUS316 Stainless Steel
Circulator Nozzles	2 for Primary Prism, 2 for Secondary Prism; accepts 3/8" ID tubing
Dimensions *(L x W x H in/mm)*	6.5 x 3.5 x 9.4 / 165.1 x 88.9 x 238.76
Weight	9.9 lb / 4.5 kg
Supplied With	2 x LR44, 1.5V Batteries
	Glass Standard (1.5163 nD)
	Bromonaphthalene Contact Liquid (4 mL bottle)
	Scale Adjustment Tool
Warranty	1-Year Parts and Labor



C10

C10 Abbe Refractometer

Item No.	Scale	Range	Resolution	Accuracy
C10	Brix	0.0 to 95.0%	0.25% Brix	±0.15% Brix
	Refractive Index	1.3000 to 1.7200 nD	0.0005 nD	±0.00025 nD

Refractometer Standards, Letter of NIST Traceability Certification

Item No.	Type	Brix Value (68°F / 20°C)	R.I. Value (68°F / 20°C)	Bottle Volume
45000-OS025	Oil	60.00%	1.4420 nD	7.4 mL / 0.25 oz
45000-OS200	Water	0.00%	1.33297 nD	7.4 mL / 0.25 oz
45000-OS001	Contact Solution	0.00%	1.65 nD	7.4 mL / 0.25 oz

Manual Polarimeter

Rugged design and simple operation allows for heavy use at the instructional level

- Polarimeters are used to measure the purity of ingredients in pharmaceuticals, quality control of raw materials, and concentration of ingredients in sugar-based substances
- The model C25L Polarimeter is designed to measure the angle of rotation caused by passing polarized light through an optically active substance
- Power Output: 110/220 VAC, 50/60 Hz. Comes with 110 VAC US power cord



Manual Polarimeter

Item No.	Measuring Range	Resolution	Accuracy	Light Source	L x W x H (in)	L x W x H (mm)	Weight	Power
4601P	±180°	0.1°	±0.1°	LED with 589 nm filter	18.1 x 5.5 x 13	460 x 140 x 330	11.5 lbs / 5.2 kg	110/220 VAC

Replacement parts

Item No.	Description
4600P-SC100	100 mm sample cell
4600P-SC200	200 mm sample cell
1200-FS4	Fuse, 1A, 250V



4600P-SC200, 200 mm, 18 mL

4600P-SC100, 100 mm, 11 mL

HYDROMETERS

VEE GEE Hydrometers are manufactured to exacting standards. Each instrument is individually tested and inspected to ensure the highest level of dependability, accuracy, and uniformity. Scales include: API, Alcohol, Battery, Baume, Brix, Calcium Chloride, Liquid Petroleum Gas, Plato, Specific Gravity, Sodium Chloride & Soil Analysis.

Hydrometers Explained . . .

The hydrometer is an instrument which is constructed on the Archimedes principle that a solid body displaces its own weight of the liquid in which it floats. Hydrometers can be divided into two general classes; namely for liquid heavier than water and for liquids lighter than water. The base hydrometer scale is Specific Gravity, in which distilled water equals 1.000 as the initial point. Liquids lighter than water are scaled below 1.000 specific gravity and liquids heavier than water are scaled above 1.000 specific gravity.

Many other scales are commonly used, such as API, Brix, Baume, Plato, etc.
All of which are convertible into specific gravity by formula.

Hydrometers are usually calibrated at 15°C/60°F. To determine the density of a liquid, the liquid should be at 15°C/60°F. If the temperature varies, the liquid will either contract or expand, depending upon the temperature. Therefore, the density fluctuates with the temperature. Where there is a variation from the standard 15°C/60°F, corrections must be applied to the hydrometer reading. To assure proper corrections, a separate accurate thermometer should be used, or a hydrometer in combination with a thermometer, which is sometimes referred to as a "thermohydrometer."



The correct method of reading a hydrometer follows:

A. Observe a point below the plane of the liquid surface. The surface should appear as an ellipse (Fig. 1).

B. The line of vision is raised until the surface, seen first as an ellipse, becomes a straight line (Fig. 2).

C. The point at which this line cuts the hydrometer scale is the reading of the instrument (Fig. 3).

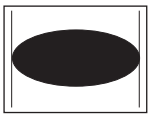

Fig. 1


Fig. 2

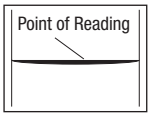


Fig. 3

If the liquid is not sufficiently clear for readings to be made in this manner, read from above the surface and estimate as accurately as possible the point to which the liquid rises on the hydrometer stem. Since hydrometers are calibrated to give correct indications when read at the principal surface of the liquid, correct the reading just taken at the upper edge of the meniscus by an amount equal to this height above the principal surface of the liquid. The amount of correction can be determined with sufficient accuracy for most purposes by taking a few readings on the upper and lower meniscus in a clear liquid of the same character as that being tested and noting the differences.

Alcohol Hydrometers - Tralle and Proof Scales

Verify the alcohol content of distilled spirits

Both models include the same ranges; model 6612-2TS also features an internal thermometer with SafetyBLUE fill for measuring solution temperature from -10 to +100°F.

- Calibrated for use at 15°C/60°F

NOTE: **Not for use with beer or wine.**

Alcohol Hydrometers - Tralle and Proof Scales

Item No.	Range	Subdivisions/ Tolerance	Range	Subdivisions/ Tolerance	Approx. Length *(in/mm)*
	TRALLE		PROOF		
6612-1	0 to 100%	1% / ±1%	0 to 200	2 / ±2%	12 / 305
6612-2TS	0 to 100%	1% / ±1%	0 to 200	2 / ±2%	14 / 355



6612-2TS

Alcohol Hydrometers - Proof Scale, Internal Revenue Service Spec.

Verify the alcohol content of your ethyl alcohol-based spirits with these hydrometers

- Tolerances match those laid out by the Internal Revenue Service specifications
- Calibrated for use at 15°C/60°F

Alcohol Hydrometers - IRS Specification

Item No.	Proof Range	IRS Size	Subdivisions	Tolerance	Approx. Length *(in/mm)*
6613-B	80 to 120	B	0.5	±0.5%	9 / 230
6613-F	0 to 20	F	0.2	±0.3%	12 / 305
6613-G	20 to 40	G	0.2	±0.4%	12 / 305
6613-H	40 to 60	H	0.2	±0.4%	12 / 305
6613-I	60 to 80	I	0.2	±0.3%	12 / 305
6613-K	75 to 95	K	0.2	±0.3%	12 / 305
6613-L	90 to 110	L	0.2	±0.3%	12 / 305
6613-M	105 to 125	M	0.2	±0.3%	12 / 305
6613-N	125 to 145	N	0.2	±0.3%	12 / 305
6613-P	145 to 165	P	0.2	±0.3%	12 / 305
6613-Q	165 to 185	Q	0.2	±0.2%	12 / 305
6613-R	185 to 206	R	0.2	±0.2%	12 / 305



6613-F

Alcohol Hydrometers w/3-point NIST Certificate and PVC Case- Proof Scale, Internal Revenue Service Specification

Verify the alcohol content of your ethyl alcohol-based spirits with these hydrometers

Tolerances match those laid out by the Internal Revenue Service specifications.

- Commonly used in distilleries requiring certification
- Calibrated for use at 15°C/60°F

INCLUDES: Hydrometer, NIST-traceable certificate with 3 data points, and a protective PVC case

Alcohol Hydrometers w/ NIST Cert. IRS Specs. & Case

Item No.	Proof Range	IRS Size	Subdivisions	Tolerance	Approx. Length *(in/mm)*
6613-B-C	80 to 120	B	0.5	±0.5%	9 / 230
6613-F-C	0 to 20	F	0.2	±0.3%	12 / 305
6613-G-C	20 to 40	G	0.2	±0.4%	12 / 305
6613-H-C	40 to 60	H	0.2	±0.4%	12 / 305
6613-I-C	60 to 80	I	0.2	±0.3%	12 / 305
6613-K-C	75 to 95	K	0.2	±0.3%	12 / 305
6613-L-C	90 to 110	L	0.2	±0.3%	12 / 305
6613-M-C	105 to 125	M	0.2	±0.3%	12 / 305
6613-N-C	125 to 145	N	0.2	±0.3%	12 / 305
6613-P-C	145 to 165	P	0.2	±0.3%	12 / 305
6613-Q-C	165 to 185	Q	0.2	±0.2%	12 / 305
6613-R-C	185 to 206	R	0.2	±0.2%	12 / 305



6613-F-C

Plato Hydrometers with Thermometers

Calibrated to show the concentration of extract as a percentage by weight in brewery wort

For example, a 5°P reading indicates 5 g of extract per 100 g of wort.

- Primarily used in the brewing industry
- Calibrated for use at 68°F/20°C

Plato Hydrometers with Thermometers

Item No.	Plato Range	Subdivisions	Approx. Length *(in/mm)*	Thermometer Range	Thermometer Subdivision	Thermometer Liquid Fill
6614TS-5	0 to 8.5°	0.1°	14 / 360	0 to 50°C	1°C	SafetyBLUE
6614TS-6	7.5 to 16°	0.1°	14 / 360	0 to 50°C	1°C	SafetyBLUE
6614TS-7	15.5 to 24°	0.1°	14 / 360	0 to 50°C	1°C	SafetyBLUE
6614TS-8	0 to 32°	0.5°	14 / 380	30 to 120°F	2°F	SafetyBLUE



6614TS-5

API ASTM Hydrometers with Thermometer

Chosen as the Standard Scale for Petroleum

The API Scale was selected as the standard hydrometer scale for petroleum products by the American Petroleum Institute (API), the U.S. Bureau of Mines, and the National Institute of Standards and Technology (NIST).

- The API Scale is based on the following formula: °API @ 60°F = (141.5/(Specific Gravity at 60°F) - 131.5
- The thermometer scale is located in the body of the hydrometer
- Calibrated for use at 15°C/60°F

API ASTM Hydrometers with Thermometer

Item No.	ASTM No.	API Range	Subdivisions	Approx. Length *(in/mm)*	Thermometer Range	Thermometer Subdivision	Thermometer Liquid Fill
6752HTS	52H	9 to 21°	0.1°	15 / 380	0 to 150°F	2°F	SafetyBLUE
6753HTS	53H	19 to 31°	0.1°	15 / 380			
6754HTS	54H	29 to 41°	0.1°	15 / 380			
6755HTS	55H	39 to 51°	0.1°	15 / 380			
6756HTS	56H	49 to 61°	0.1°	15 / 380			
6757HTS	57H	59 to 71°	0.1°	15 / 380			



6756HTS

Battery Hydrometer - Syphon Set

For Measuring Battery Acid

The Battery Acid Hydrometer Set is used in many industrial settings including, fleet maintenance, automotive services, solar-powered system maintenance, and general industrial applications.

- The rubber parts are made of nitrile rubber, which is chemically inert and durable
- A reinforced box provides storage and is compartmented to eliminate breakage
- Calibrated for use at 25°C/77°F

INCLUDES: Specific Gravity Hydrometer, glass syphon tube with rubber bulb, nozzle, and clamp

Battery Hydrometers

Item No.	Description	Specific Gravity Range	Subdivisions	Approx. Length *(in/mm)*
6605-5	Hydrometer and Syphon Set	1.150 to 1.300	0.005	20 / 507
6605-5H	Hydrometer Only	1.150 to 1.300	0.005	8.7 / 220
6605-OSY	Glass syphon tube, rubber bulb, nozzle, and clamp only	-------	-------	20 / 507



6605-5

Baume (Heavy) Hydrometers

These Baume Hydrometers are for liquids heavier than water

- The Baume scale is based on the following formula: Specific Gravity = 145/(145-degrees Baume)
- Calibrated for use at 15°C/60°F

Baume (Heavy) Hydrometers

Item No.	Baume Range	Subdivisions	Length *(in/mm)*
6609-1	0 to 12°	0.1°	12 / 305
6609-2	9 to 21°	0.1°	
6609-3	19 to 31°	0.1°	
6609-4	29 to 41°	0.1°	
6609-5	39 to 51°	0.1°	
6609-8	0 to 15°	0.1°	
6609-9	0 to 25°	0.2°	

Baume (Heavy) Hydrometers

Item No.	Baume Range	Subdivisions	Length *(in/mm)*
6609-10	0 to 35°	0.5°	12 / 305
6609-11	0 to 50°	0.5°	
6609-12	35 to 70°	0.5°	
6609-13	0 to 50°	1.0°	
6609-14	0 to 70°	1.0°	
6609-15	0 to 90°	1.0°	



6609-1